Code No: 124AG **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, April/May - 2023
FORMAL LANGUAGES AND AUTOMATA THEORY
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a) What are Moore and Melay Machines? [2]
b) Give the general procedure for drawing a state diagram from the transition table. [3]
c) Represent the language that contains strings over $\sum = \{0,1\}$ and has even number of 0's. [2]
d) State and define Kleene's theorem. [3]
e) Write down the formal statement of pumping lemma. [2]
f) Give the CFG for the following language: $L = \{0^n 1^m 0^{n+m} \mid n \geq 0, m \geq 0\}$. [3]
g) What is a Turing Machine? [2]
h) Design a TM to find the value of $\log_2(n)$, where n is any binary number and a perfect power of 2. [3]
i) What is meant by the Halting problem? [2]
j) Discuss about various types of grammars. [3]

PART - B

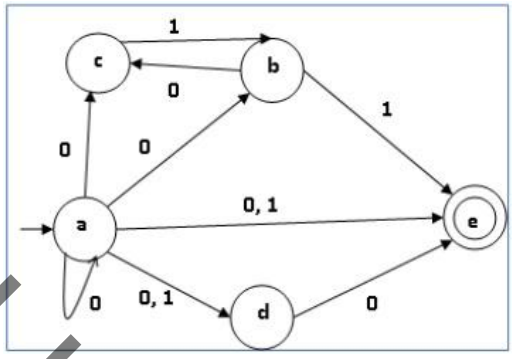
(50 Marks)

2.a) Write down the steps in converting NFA to DFA.
b) Convert the NFA: N=[{p,q,r,s},{0,1},δ,p,{s}], where δ is given in the table below to its equivalent DFA. [4+6]

Q \ Σ	0	1
p	p,q	p
q	r	r
r	s	-
s	s	s

OR

3.a) Minimize the following DFA.



b) Design a Moore Machine to determine the residue mod 4 for each binary string treated as integer. [5+5]

4.a) Show that (a*b*)* = (a+b)*.

b) Construct the NFA for the regular expression r = ((0+1)*(00)*(11)*. [4+6]

OR

5.a) Using a regular expression represent the set of all strings of a's and b's containing at least one combination of double letters.

b) List and explain the closure properties of regular sets with examples. [4+6]

6. Explain in detail about Greiback normal form with suitable example. [10]

OR

7.a) Construct the PDA to the following grammar:

S→AB
A→BS/b
B→SA/a

b) List the properties of a context-free language and explain. [6+4]

8. Design a T.M for copying of information from one place to the other place. Make all the necessary assumptions. Give example of the working of your T.M. [10]

OR

9.a) Design a Turing Machine that accepts the language $\{0^n1^n0^n \mid n \geq 1\}$.

b) Design a TM that recognized binary palindrome. [5+5]

10. Describe in detail about undecidability of Post's Correspondence Problem. [10]

OR

11.a) Discuss briefly about decidability and undecidability problems.

b) Explain in detail about Turing reducibility. [5+5]

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